

Assignment 1

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Assignment I

Subject : Automata Theory

Topic : Deterministic Finite Automata (DFA)

* Question 1

Construct a DFA that accepts all strings having an even number of 1's over the alphabet $\{0, 1\}$

• Aim :

To construct a Deterministic Finite Automata (DFA) that accepts strings containing an even number of 1's.

• Theory :

A Deterministic Finite Automata (DFA) is an automata in which for each state and input symbol, exactly one transition is defined.

In this problem, the DFA tracks whether the number of 1's is even or odd.

• DFA Description :

- states (Q) : $\{q_0, q_1\}$
- Alphabet (Σ) : $\{0, 1\}$
- Start state : $\{q_0\}$
- Final state : $\{q_0\}$

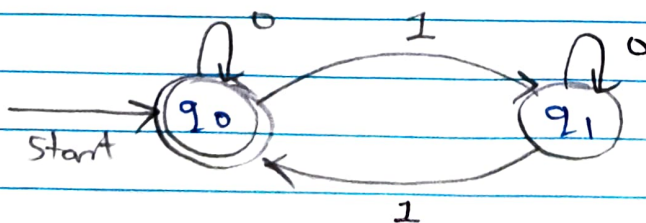
• State Meaning :

- q_0 : Even number of 1's
- q_1 : Odd number of 1's

• Transition Table :

		Symbols	
		0	1
states	q_0	q_0	q_1
	q_1	q_1	q_0

• DFA Diagram :



Question 2:

Construct a DFA that accepts all strings having an even no. of 0's and even no. of 1's over the alphabet $\{0,1\}$

• Aim:

To construct a DFA that accepts strings in which both 0's and 1's occur an even number no. of times.

Theory:

To check the parity (even or odd) of both input symbols, the DFA uses four states.

Each state represents a different combination of even and odd no. of 0's and 1's.

• DFA Description:

- States (Q): $\{q_0, q_1, q_2, q_3\}$
- Alphabets (Σ): $\{0,1\}$
- Start state: q_0
- Final state: $\{q_0\}$

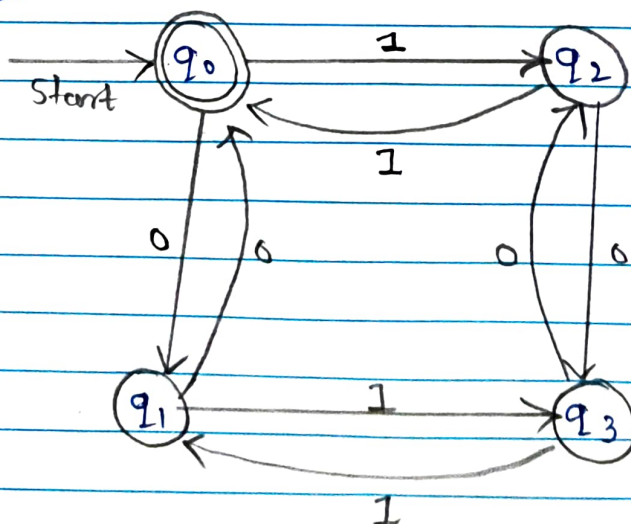
State Meanings :

- q_0 : Even 0's , Even 1's
- q_1 : Odd 0's , Even 1's
- q_2 : Even 0's , Odd 1's
- q_3 : Odd 0's , Odd 1's

Transition Table :

State \ Symbol	0	1
q_0	q_1	q_2
q_1	q_0	q_3
q_2	q_3	q_0
q_3	q_2	q_1

DFA Diagram :



• Simulation :

Let's consider a string "1010".

$\delta(q_0, 1)$

$\delta(q_2, 0)$

$\delta(q_3, 1)$

$\delta(q_1, 0) = q_0$ (Final state)

As it ends at q_0 (accepting string) string is accepted.

• Conclusion :

In this assignment, DFA's were constructed to:

1. Accept string with an even number of 1's
2. Accept strings with an even number of both 0's and 1's.

Both automata correctly recognize the required language over the alphabet $\{0, 1\}$.

Code :

```
#include <stdio.h>

#include <string.h>

int main() {
    int states, final_state, start_state;
    int transition[10][2]; // Max 10 states, 2 symbols (0 and 1)
    char input[100];

    printf("Enter number of states: ");
    scanf("%d", &states);

    printf("Enter transition table (for 0 and 1):\n");
    for(int i = 0; i < states; i++) {
        printf("For state q%d:\n", i);
        printf("  On 0 -> ");
        scanf("%d", &transition[i][0]);
        printf("  On 1 -> ");
        scanf("%d", &transition[i][1]);
    }

    printf("Enter start state: ");
    scanf("%d", &start_state);

    printf("Enter final state: ");
    scanf("%d", &final_state);

    printf("Enter input string (only 0 and 1): ");
    scanf("%s", input);
```



```

int current = start_state;

printf("\nTransition Details:\n");

for(int i = 0; i < strlen(input); i++) {
    int symbol = input[i] - '0'; // convert char to int
    printf("q%d --%c--> q%d\n", current, input[i], transition[current][symbol]);
    current = transition[current][symbol];
}

if(current == final_state)
    printf("\nString Accepted\n");
else
    printf("\nString Rejected\n");

return 0;
}

```

Output:

```

Enter number of states: 2
Enter transition table (for 0 and 1):
For state q0:
    On 0 -> 0
    On 1 -> 1
For state q1:
    On 0 -> 0
    On 1 -> 1
Enter start state: 0
Enter final state: 1
Enter input string (only 0 and 1): 1011

Transition Details:
q0 --1--> q1
q1 --0--> q0
q0 --1--> q1
q1 --1--> q1

String Accepted

```